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Precision of 7 commercially available devices for predicting the bench press 1-repetition maximum from the individual load-velocity relationship

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ABSTRACT

Objective: This study aimed to compare the accuracy of different devices to predict the bench press 1-repetition maximum (1RM) from the individual load-velocity relationship modelled through the multiple- and 2-point methods. ***Methods:*** 11 men performed an incremental test on Smith machine against 5 loads (45–55–65–75–85%1RM), followed by 1RM attempts. The mean velocity was simultaneously measured by 1 linear velocity transducer (T-Force), 2 linear position transducers (Chronojump and Speed4Lift), 1 camera-based optoelectronic system (Velowin), 2 inertial measurement units (PUSH band and Beast sensor), and 1 smartphone application (PowerLift). The velocity recorded at the 5 loads (45–55–65–75–85%1RM) or only at the 2 most distant loads (45-85%1RM) were considered for the multiple- and 2-point methods, respectively. ***Results:*** An acceptable and comparable accuracy in the estimation of the 1RM was observed for the T-Force, Chronojump, Speed4Lift, Velowin and PowerLift when using both the multiple- and 2-point methods (effect size [ES] ≤ 0.40; Pearson's correlation coefficient [*r*] ≥ 0.94; standard error of the estimate [SEE] ≤ 4.46 kg), while the accuracy of the PUSH (ES = 0.70-0.83; *r* = 0.93-0.94; SEE = 4.45-4.80 kg) and especially the Beast sensor (ES = 0.36-0.84; *r* = 0.50-0.68; SEE = 9.44-11.2 kg) was lower. ***Conclusions:*** These results highlight that the accuracy of 1RM prediction methods based on movement velocity are device-dependent with the inertial measurement units providing the least accurate estimate of the 1RM.

Keywords: maximum dynamic strength, linear position transducer, camera-based optoelectronic system, inertial measurement units, smartphone application.

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INTRODUCTION

Movement velocity has been proposed as a non-fatiguing method of predicting the 1RM.[1] An increasing number of devices are available in the market to measure movement velocity. The linear position and velocity transducers have been the 2 most commonly used devices to examine the velocity-based resistance training approach.[2] However, the interest in wearable wireless technology (camera-based optoelectronic device, inertial measurement units, or smartphone applications) is growing.[3–5] However, conflicting results exist regarding the validity of wearable wireless devices to measure movement velocity.[5–7] A practical question that remains virtually unexplored is whether wearable wireless devices could predict a 1RM with an accuracy comparable to the most commonly used linear position and velocity transducers.

Recent studies have recommended the individual load-velocity relationship for obtaining a more accurate prediction of the 1RM.[8] The recording of mean velocity values against 2 or more submaximal loads allows estimating the 1RM through a linear regression as the load associated with the velocity of the 1RM.[8,9] In this regard, it would be important to explore the number of loads needed for obtaining an accurate prediction of the 1RM. To date, no study has compared the accuracy in the prediction of the 1RM between the multiple- (more than 2 loads) and 2-point (only 2 loads) methods.

The aim of this study was to explore the accuracy of 7 commercially available devices (1 linear velocity transducer, 2 linear position transducers, 1 camera-based optoelectronic device, 2 inertial measurement units, and 1 smartphone application) to predict the 1RM from the individual load-velocity relationship modelled by 5 (multiple-point method) or 2 (2-point method) loads during the concentric-only bench press exercise performed in a Smith machine. It was hypothesized that (I) the linear position and velocity transducers would provide a more

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accurate prediction of the 1RM than wearable wireless devices,[7] and (II) the accuracy in the estimation of the 1RM would be comparable for the multiple- and 2-point methods.

METHODS

Subjects

Eleven men (age: 22.5 ± 1.9 years; body mass: 75.2 ± 7.2 kg; height: 1.75 ± 0.06 m; bench press 1RM: 83.8 ± 12.3 kg [1RM relative to body mass: 1.10 ± 0.16 kg·kg^{-1}]) experienced with the bench press exercise participated in this study. Subjects signed a written informed consent form prior to initiating the study. The study protocol adhered to the tenets of the Declaration of Helsinki and was approved by the University of Granada Institutional Review Board.

Testing procedures

Subjects performed 2 testing sessions separated by 48-72 h. The first session was used for anthropometric measures and to determine the bench press 1RM. The warm-up consisted of jogging, joint mobilization exercises, and 1 set of 5 repetitions against an external load of 37 kg during the bench press exercise. Thereafter, the load was incremented by 10 to 1 kg until the 1RM load was reached. The inter-set rest was set to 4 min and 1-2 repetitions were performed with each load. The second session began with the same warm-up described above. Subjects then performed 3 repetitions against 5 incremental loads (45–55–65–75–85%1RM), followed by 1RM attempts (a maximum of 3 attempts were allowed). The inter-repetition and inter-set rest were set to 15 s and 4 min, respectively. The barbell was held by the safety stops of the Smith machine 1-2 cm above subjects' chest at the level of the sternum. From that position, subjects were instructed to perform a concentric-only movement as fast as possible until their elbows reached full extension. The actual 1RM value was highly reproducible

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(coefficient of variation = 2.05%; intraclass correlation coefficient = 0.98). The 1RM determined in the second session was used for the validity analysis.

The bench press was performed in a Smith machine (Technogym, Barcelona, Spain). The mean velocity data was simultaneously monitored by 7 commercial devices: 1 linear velocity transducer (T-Force [v.2.28, T-Force System, Ergotech, Murcia, Spain]), 2 linear position transducers (Chronojump [v.1.6.2, Chronojump Boscosystem®, Barcelona, Spain] and Speed4Lift [v.4.1, Speed4Lift, Madrid, Spain]), 1 camera-based optoelectronic device (Velowin [v.1.6.314, Velowin, DeporTeC, Murcia, Spain]), 2 inertial measurement units (PUSH band [v1.1.26, PUSHTH band, PUSH Inc., Toronto, Canada] and Beast sensor [v.2.3.7, Beast sensor, Beast Technologies Srl., Brescia, Italy]), and 1 smartphone application (PowerLift) (Figure 1).

Only the repetition with the highest mean velocity for each loading condition and device was used for subsequent analysis. The individual load-velocity relationship was modelled using the data of 5 (45–55–65–75–85%1RM; "multiple-point method") and only 2 loads (45–85%1RM; "2-point method"). Due to the low reliability of the velocity of the 1RM and the trivial differences between the between- and within-subject variability for the velocity of the 1RM,[8,9] the 1RM was estimated from the individual load-velocity relationship as the load associated with a standard mean velocity of 0.17 m·s^{-1}.[8]

Statistical analyses

The accuracy of the 1RM estimated from the 7 devices and 2 methods respect to the actual 1RM was examined through paired samples *t*-tests, the Hedge's *g* effect size (ES),[10] the Pearson's correlation coefficient (*r*) and the standard error of the estimate (SEE). The magnitude of the ES was interpreted as follows: *trivial* (< 0.2), *small* (0.2-0.59), *moderate* (0.60-1.19), *large* (1.2-2.0) and *very large* (> 2.0).[11] The strength of the *r* coefficients was

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interpreted as follows: *trivial* (< 0.1), *small* (0.1-0.3), *moderate* (0.3-0.5), *high* (0.5-0.7), *very high* (0.7-0.9) or *practically perfect* (> 0.9).[11] The devices were deemed highly valid if the following criteria were met: trivial to small ES, practically perfect *r*, and moderate SEE (< 5 kg). A two-way repeated-measured ANOVA (device × method) with Bonferroni post-hoc corrections was applied on the absolute differences between the actual and predicted 1RM. The Greenhouse-Geisser correction was used when the Mauchly's sphericity test was violated. Statistical analyses were performed using the software package SPSS (version 22.0: SPSS, Ins., Chicago, IL, USA). Alpha was set at $p < 0.05$.

RESULTS

An acceptable and comparable accuracy in the estimation of the 1RM was observed for the T-Force, Chronojump, Speed4Lift, Velowin and PowerLift when using both the multiple- and 2-point methods (ES ≤ 0.40; $r \geq 0.94$; SEE ≤ 4.46 kg), while the accuracy of the PUSH band (ES = 0.70-0.83; r = 0.93-0.94; SEE = 4.45-4.80 kg) and especially the Beast sensor (ES = 0.36-0.84; r = 0.50-0.68; SEE = 9.44-11.2 kg) was lower (Table 1).

The absolute differences between the actual and predicted 1RM revealed a significant main effect for device ($F_{2.2, 21.9} = 15.3$, $p < 0.001$), but not for the method ($F_{1,10} = 0.28$, $p = 0.609$) or their interaction ($F_{2.2, 22.2} = 2.84$, $p = 0.075$). The absolute differences were larger for the Beast sensor and PUSH band compared to the other devices (Figure 2).

DISCUSSION

There are different devices available in the market for monitoring movement velocity in real-time during resistance training exercises.[5,7] Since the successive manipulation of the raw data can magnify the measurement error of mean velocity,[6,7] it was hypothesised that the linear position and velocity transducers would provide a more accurate prediction of the 1RM. This hypothesis was confirmed with respect to the two inertial measurement units (PUSH band

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and especially the Beast sensor), but no meaningful differences were observed compared to the camera-based optoelectronic device (Velowin) and the smartphone application (PowerLift). The Velowin was equally accurate when compared to linear position and velocity transducers, while Powerlift presented a higher absolute error of only ≈ 1.5 kg. Conversely, the PUSH band significantly underestimated the 1RM and it presented larger absolute errors compared to the other devices (with the exception of the Beast sensor), while the Beast sensor provided the least accurate estimate of the 1RM. Previous studies exploring the validity of the wearable wireless devices examined in the present study to monitor movement velocity under individual loads have revealed a high validity for Velowin (*r* range: 0.97 to 0.98),[3] Powerlift (*r* range: 0.94 to 0.98),[4,5] and Beast sensor (*r* range: 0.94 to 0.98),[5] while conflicting results have been reported for the PUSH band (*r* range: 0.30-0.85).[6,7] The results of the present study suggest that the accuracy of inertial measurement units is not still good enough to provide an accurate estimation of the 1RM through the individual load-velocity relationship.

The 2-point method has been recently proposed as a quick and fatigue-free procedure for assessing the 1RM.[12] Garcia-Ramos et al.[8] reported a high reliability (coefficient of variation ≤ 5.11%) and practically perfect correlations ($r \geq 0.94$) for the 1RM estimated through the 2-point method in the bench press exercise. However, this is the first study that has explicitly compared the accuracy in the estimation of the 1RM between the 2-point method and the most commonly used multiple-point method. Our findings revealed a comparable precision in the estimation of the 1RM between both methods.

PRACTICAL APPLICATIONS

The 1RM during the bench press exercise performed in a Smith machine can be accurately estimated from the velocity recorded under only 2 loads with the T-Force, Chronojump, Speed4Lift, Velowin and PowerLift. The application of the 2-point method to

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estimate the bench press 1RM requires 2 steps: (1) recording of the mean velocity against 2 loads representing approximately the 45%1RM and 85%1RM, (2) modelling of the individual load-velocity relationship and determining the 1RM as the load associated with a mean velocity of 0.17 $m \cdot s^{-1}$.

CONCLUSIONS

The linear velocity transducer (T-Force), 2 linear position transducers (Chronojump and Speed4Lift), 1 camera-based optoelectronic system (Velowin) and 1 smartphone application (PowerLift) provided the 1RM value with an acceptable and comparable precision. The accuracy in the estimation of the 1RM by the 2 inertial measurement units (PUSH band and especially the Beast sensor) was significantly lower. Moreover, the precision in the estimation of the 1RM did not differ between the multiple-point (45–55–65–75–85% of 1RM) and 2-point (45–85% of 1RM) methods. These results question the feasibility of the inertial measurement units to estimate the bench press 1RM through the individual load-velocity relationship, while they encourage the use of the 2-point method as a quicker, less prone to fatigue and equally accurate procedure of estimating the 1RM during the bench press exercise performed in a Smith machine.

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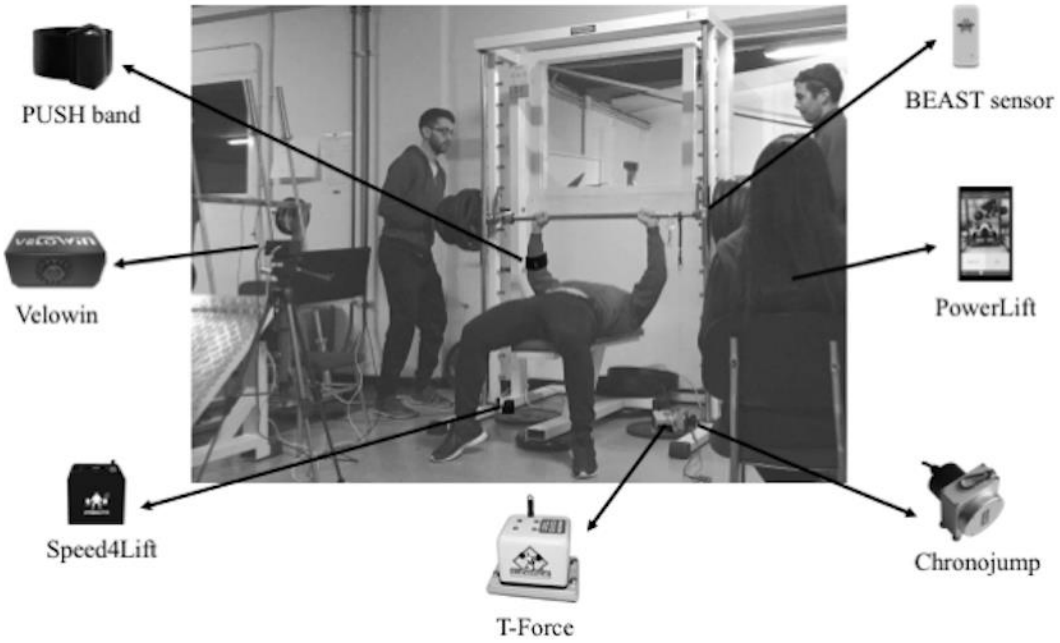

Figure 1. Distribution of the different devices during the bench press testing protocol.

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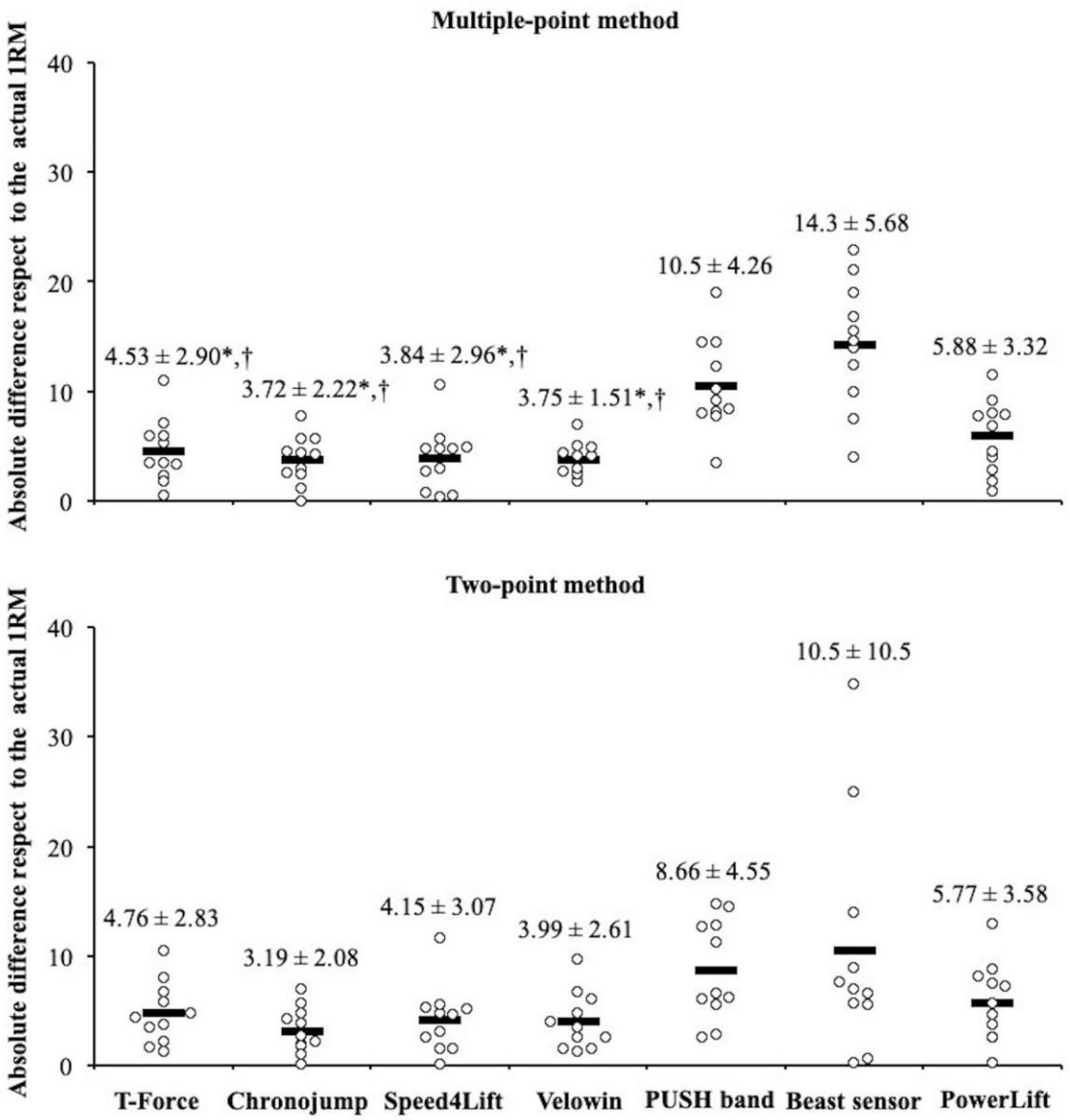

Figure 2. Comparison of the absolute differences between the actual 1-repetition maximum (1RM) and the 1RM estimated by means of 7 different commercially available devices using the multiple- (upper panel) and 2-point (lower panel) methods. Means ± standard deviations are indicated. *, significant differences respect to PUSH band; †, significant differences respect to Beast sensor ($p < 0.05$ with Bonferroni correction).

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Table 1. Comparison of the actual bench press 1-repetition maximum (1RM) vs. the 1RM estimated by means of 7 different commercially available devices using the multiple- and 2-point methods.

Method	Device	Estimated 1RM (mean ± SD)	*p*	ES (95% CL)	*r* (95% CL)	SEE (95% CL)
Multiple-point	T-Force	89.3 ± 12.4	0.001	0.35 (-1.19, 0.49)	0.97 (0.91, 0.99)	3.16 (2.30, 5.20)
	Chronojump	83.1 ± 12.9	.0195	-0.13 (-0.71, 0.97)	0.95 (0.84, 0.98)	4.11 (3.00, 6.76)
	Speed4Lift	88.7 ± 12.6	0.002	0.30 (-1.14, 0.54)	0.97 (0.91, 0.99)	3.02 (2.21, 4.98)
	Velowin	87.2 ± 13.5	0.045	0.18 (-1.02, 0.66)	0.97 (0.91, 0.99)	3.15 (2.30, 5.18)
	PUSH band	74.4 ± 12.0	< 0.001	-0.83 (-0.04, 1.70)	0.94 (0.82, 0.98)	4.45 (3.25, 7.32)
	Beast sensor	73.4 ± 14.0	0.005	-0.84 (-0.04, 1.71)	0.68 (0.24, 0.89)	9.44 (6.89, 15.5)
	PowerLift	89.8 ± 13.4	0.005	0.37 (-1.22, 0.47)	0.94 (0.82, 0.98)	4.46 (3.25, 7.33)
2-point	T-Force	89.4 ± 13.1	0.001	0.35 (-1.19, 0.49)	0.97 (0.91, 0.99)	3.12 (2.27, 5.13)
	Chronojump	83.8 ± 13.2	0.372	-0.08 (-0.76, 0.91)	0.96 (0.87, 0.99)	3.68 (2.68, 6.05)
	Speed4Lift	88.9 ± 12.8	0.001	0.31 (-1.15, 0.53)	0.97 (0.91, 0.99)	3.13 (2.28, 5.15)
	Velowin	88.1 ± 13.6	0.013	0.24 (-1.08, 0.59)	0.97 (0.90, 0.99)	3.27 (2.39, 5.38)
	PUSH band	76.2 ± 11.3	< 0.001	-0.70 (-0.16, 1.57)	0.93 (0.79, 0.98)	4.80 (3.50, 7.90)
	Beast sensor	90.0 ± 15.6	0.258	0.36 (-1.20, 0.49)	0.50 (-0.03, 0.81)	11.2 (8.15, 18.4)
	PowerLift	90.1 ±13.3	0.002	0.40 (-1.24, 0.45)	0.95 (0.84, 0.98)	4.13 (3.02, 6.80)

p, *p*-value; ES, Hedge's *g* effect size; *r*, Pearson's correlation coefficient; SEE, standard error of the estimate; SD, standard deviation; 95% CL, 95% confidence limit. The actual 1RM value was 83.8 ± 12.3 kg.

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